Nguyen Thach Thanh

Embedded Systems & IoT Engineering

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Final-year Microelectronics Engineering student at HUFLIT, working where hardware meets firmware: I designed a 4-layer low-power PCB, wrote bare-metal STM32 firmware, and measured input energy of a duty-cycled IoT node across 286 instrumented runs. Co-author of two international conference papers (Springer CCIS; CUTE 2026). Looking for an embedded systems internship.

EDUCATION

Ho Chi Minh City University of Foreign Languages – Information Technology (HUFLIT) 2023 – Present

B.Sc. Information Technology | Major: Microelectronics Engineering

PUBLICATIONS

Efficient FPGA-Based Hardware Noise Filtering and Real-Time Motion Detection Using RE200B Sensor on Trion T8 BGA81 Springer CCIS, 2026

Co-author — Computational Intelligence in Engineering Science, *doi:10.1007/978-3-031-98167-8_21*

Toward a Hybrid Efficient Edge-Host Smart Helmet System for Drowsiness Detection and Accident Verification using ESP32-S3 CUTE 2026

Co-author — The 5th International Conference on Sustainable Development in Civil, Urban and Transportation Engineering

PROJECTS

Buck vs LDO Energy Comparison 01/2026 – 08/2026

- Designed and brought up a 4-layer KiCad PCB with jumper-selectable TPS62840 buck / STLQ020 LDO branches powering an STM32L432 + BME280 node, and wrote the bare-metal firmware for the 5 s BME280–Stop2 duty cycle (race-to-sleep).
- Built an INA228 input-side energy rig (∼10k samples/s) and ran 286 verified experiments across 3.30–4.20 V — each run as a statistical unit, with confidence intervals plus ABBA and marker verification campaigns.
- Showed the buck-vs-LDO trade-off is voltage-region-dependent: buck draws 10–18% less across 3.80–4.20 V, but enters a high-input-energy state in the 3.35–3.60 V region (up to ∼75× the LDO).

GreenEco — Smart Greenhouse 2025

- Raspberry Pi greenhouse system: a REST API driving 4 active-low relay channels (pump, two fans, grow light), with sensor data streaming to the team's server and control app.
- Brought up the hardware and sensor layer on two buses — RS485/Modbus RTU (ES-Soil7 soil, DFRobot environmental sensor) and UART (SEN0220 CO_2, SEN0501) — plus relay GPIO mapping, with Python unit tests covering the GPIO, UART/Modbus and API layers.

Smart Helmet 2025

- ESP32-S3 dual-core FreeRTOS firmware: 100 Hz MPU6050 sensing task with a filtered, speed/road-adaptive threshold fall-detection state machine (3.5 g final threshold, staged alerts, physical cancel button).
- Automatic SMS alert via GSM module (single send per crash, 10 s cancel window) and an OLED HUD fed by BLE navigation data from the phone.
- I built the sensing and firmware parts; AI drowsiness detection belonged to teammates — the work was published as a co-authored paper at CUTE 2026 (see Publications).

AWARDS

Top 12: Green Innovation Contest (Sang Tao Xanh), HUST, 2025 — GreenEco project

Top 20: BK Innovation, HCMUT, 2025 — Smart Helmet project

University-level: Research project, HUFLIT, 2025 — Smart Helmet

SKILLS

Programming: C/C++, Python, Verilog

MCU: STM32, ESP32

Protocols: SPI, I2C, UART, RS485 (Modbus RTU), BLE

Tools: KiCad, Raspberry Pi, Git, GitHub, Linux

LANGUAGES

Vietnamese (native) | English (intermediate)